Validating a Direct Survey-Based Population Synthesis Approach for Small Area Estimation of Health Indicators

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Introduction and Background

- **Small area estimation (SAE)** is a statistical technique used to analyze health outcomes and characteristics in small geographic areas.
- The **Centers for Disease Control and Prevention (CDC)** provides a widely accepted standard for generating SAE across multiple geographic scales, time periods, and health indicators using a multilevel regression approach. *However, the CDC's methods are complex and not readily available for adaptation or reuse.*
- The estimates, referred to as CDC PLACES, are externally validated using accuracy metrics such as descriptive statistics, Pearson's r, R^2, and spatial heterogeneity (Zhang et al., 2015).
- **To address this gap, this study proposes an open, reproducible approach that directly integrates public health survey data into the widely used iterative proportional fitting (IPF) technique to generate small area estimates for public health applications.**

Methods

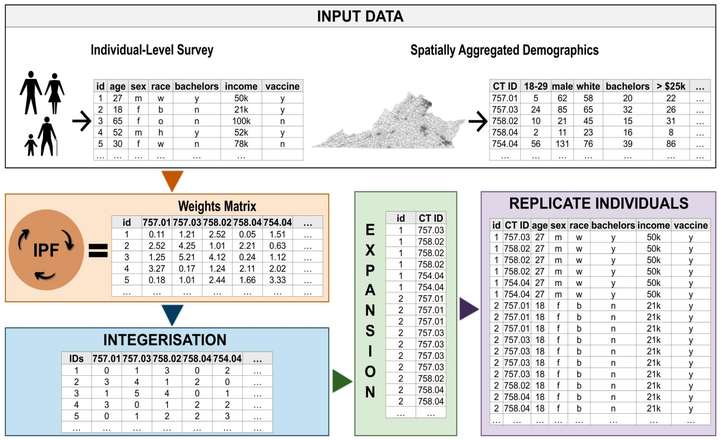


Figure 1. Overview of the direct survey-based population synthesis method for generating Small Area Estimates (SAEs) of health measures, as proposed by Von Hoene et al. (2025).

- We used the methodology described in Von Hoene et al. (2025), which generates synthetic populations using **iterative proportional fitting (IPF)** — a statistical technique that adjusts individual-level sample data to align with known population totals across multiple dimensions **(Figure 1).**
- Using this approach, we fit individual-level health survey data from the **CDC's National Health Interview Survey (NHIS)** with spatially aggregated data, including sociodemographic variables (***age, race, gender, income, education, health insurance, and urban residence***) from the U.S. Census and spatial health indicators (***obesity and smoking***) from CDC PLACES.
- Given the availability of county-level data for validation, we generated 2021 estimates for **cancer prevalence** in New York and 2019 estimates for **diabetes** and **Chronic Obstructive Pulmonary Disease (COPD)** in Florida.

Results

Overall Accuracy

Table 1. Pearson's r, R^2 and mean absolute error (MAE) comparing SAEs from the direct survey-based approach and CDC PLACES against validation datasets.

	Direct Survey-Based			CDC PLACES		
SAE	**r**	**R^2**	**MAE**	**r**	**R^2**	**MAE**
New York, 2021, Cancer	0.745	0.555	1.167	0.783	0.604	2.556
Florida, 2019, Diabetes	0.661	0.437	3.199	0.689	0.475	1.894
Florida, 2019, COPD	0.680	0.462	3.948	0.653	0.426	1.810

Descriptive Statistics

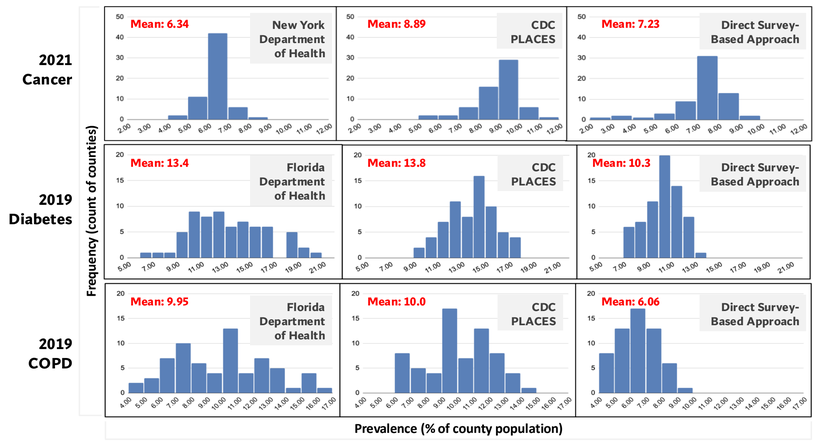


Figure 2. Comparison of frequency distributions of SAEs for cancer, diabetes, and COPD across datasets.

Spatial Heterogeneity

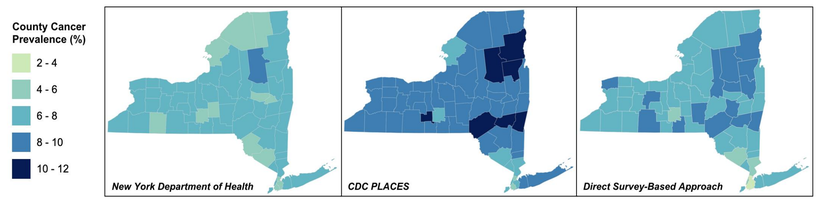


Figure 3. Comparison of 2021 county-level cancer prevalence estimates in New York.

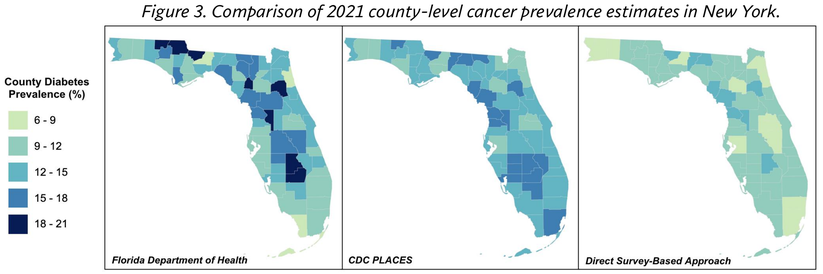


Figure 4. Comparison of 2019 county-level diabetes prevalence estimates in Florida.

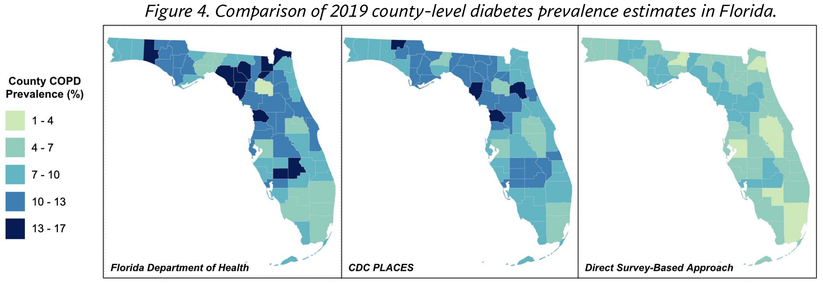


Figure 5. Comparison of 2019 county-level COPD prevalence estimates in Florida.

Discussion and Conclusions

- The direct survey-based population synthesis method was first proposed by Von Hoene et al. (2025), with later work (Von Hoene, **Gupta** et al.) demonstrating its transferability across locations, time periods, and health outcomes. **This study advances the approach by validating it against CDC PLACES estimates and establishing its utility for generating SAEs of health measures.**
- The direct survey-based approach produces SAEs of health measures that closely align with CDC PLACES in terms of:
  - **Descriptive statistics:** The direct survey-based estimates showed similar distributions to other datasets but consistently had the lowest minimum values; ranges varied across sources **(Table 1).**
  - **Overall accuracy:** Estimates from the direct survey-based approach achieved high correlation, moderate explained variance, and low error **(Table 2),** closely matching validation values.
  - **Spatial heterogeneity:** Spatial comparisons across New York and Florida counties **(Figures 2–4)** show agreement between the direct survey-based approach and observed patterns.

Advantages of the direct survey-based approach for SAE:

- **Accessibility:** This study advances SAE by offering an open and publicly available alternative for generating estimates.
- **Usability:** The method does not require complex statistical expertise, expanding access to tools that support public health research and decision-making.
- **Flexibility:** Can incorporate specialized surveys (e.g., state-based) and context-specific variables not available in CDC PLACES (e.g., attitudes, beliefs, risk perceptions), enabling more targeted and theory-informed public health insights.

References

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Von Hoene, E., Roess A., Kavak, H., & Anderson, T. (2025) Synthetic population generation with public health characteristics for spatial agent-based models. PLOS Computational Biology. https://doi.org/10.1371/journal.pcbi.1012439.

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